

Student Guide — senior / module-13- indic-ecology

IKS Curriculum

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Student Workbook

Student Workbook — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

This workbook is yours to write in. Use it for vocab, exercises, and reflection across all 10 days.

Vocabulary tracker

Fill in as we go:

- *Vasudhaiva Kuṭumbakam* — _____
- *yajña* — _____
- *ahiṃsā* — _____
- *devarakāḍu* — _____
- *sthāvara-jaṅgama* — _____

Day-by-day notes

Day 1 — Vasudhaiva Kuṭumbakam

What I learned:

One question I have:

Day 2 — Five Elements as Ecological Frame

What I learned:

One question I have:

Day 3 — Rivers as Sacred

What I learned:

One question I have:

Day 4 — Sacred Groves

What I learned:

One question I have:

Day 5 — Yajña — The Ecological Cycle

What I learned:

One question I have:

Day 6 — Pṛthivī Sūkta (AV 12.1)

What I learned:

One question I have:

Day 7 — Iṣopaniṣad — Non-Greed Ethic

What I learned:

One question I have:

Day 8 — Climate Change — The Crisis

What I learned:

One question I have:

Day 9 — From Ancient to Action

What I learned:

One question I have:

Day 10 — Project Pitch and Assessment

What I learned:

One question I have:

Final reflection

After Day 10, write a paragraph: what is the most important thing you learned in this module?

Day 1 — Vasudhaiva Kuṭumbakam

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify the source verse and explain the principle.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *Vasudhaiva Kuṭumbakam* (IAST: Vasudhaiva Kuṭumbakam; lit. “the world is one family” — Mahā Upaniṣad 6.71”)

Primary source (today’s reading)

From *Mahā Upaniṣad* 6.71–73 — *vasudhaiva kuṭumbakam* — see `sources.md`.

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see `activities/`). Today’s slice: focus on the part of the activity that reinforces Vasudhaiva Kuṭumbakam.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 2 — Five Elements as Ecological Frame

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: link module 2 *pañcabhūta* to ecology — interconnection.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *yajña* (IAST: *yajña*; lit. “cycle of giving and receiving (BG 3.10–14)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.

2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Five Elements as Ecological Frame.

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 3 — Rivers as Sacred

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: discuss the ecological function of river-worship traditions.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *ahiṃsā* (IAST: *ahiṃsā*; lit. “non-violence; extended to non-human life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Rivers as Sacred.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 4 — Sacred Groves

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: identify devarakāḍu and similar groves as biodiversity reserves.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *devarakāḍu* (IAST: devarakāḍu; lit. “sacred grove (Karnataka, S. India)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see activities/). Today’s slice: focus on the part of the activity that reinforces Sacred Groves.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.

- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 5 — Yajña — The Ecological Cycle

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: read bg 3.10–14; map yajña to natural-cycle reciprocity.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *sthāvara-jaṅgama* (IAST: sthāvara-jaṅgama; lit. “immobile (plants, rocks) and mobile (animals) life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Yajña — The Ecological Cycle.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See `homework.md` for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 6 — Pṛthivī Sūkta (AV 12.1)

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: read the earth-mother hymn and discuss imagery.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See `lesson-plan.md` for any day-specific materials)

Vocabulary introduced today

- *Vasudhaiva Kuṭumbakam* (IAST: Vasudhaiva Kuṭumbakam; lit. ““the world is one family” — Mahā Upaniṣad 6.71”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day's idea is — a number trick, a celestial pattern, a herb's identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today's idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces Pṛthivī Sūkta (AV 12.1).

Wrap-up (5 min)

- One-sentence exit ticket: “*What's one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today's task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 7 — Iṣopaniṣad — Non-Greed Ethic

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: translate *īśāvāsyam idaṁ sarvaṁ* and discuss its ecological implication.

Materials

- Whiteboard / chart paper
- Notebook, pen per student

- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *yajña* (IAST: yajña; lit. “cycle of giving and receiving (BG 3.10–14)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Iṣopaniṣad — Non-Greed Ethic.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 8 — Climate Change — The Crisis

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: summarise key ipcc findings; what is changing and why.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *ahimsā* (IAST: ahimsā; lit. “non-violence; extended to non-human life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see **activities/**). Today’s slice: focus on the part of the activity that reinforces Climate Change — The Crisis.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See homework.md for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 pañcabhūta).

Day 9 — From Ancient to Action

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: design one local environmental project applying an indic-ecology principle.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See lesson-plan.md for any day-specific materials)

Vocabulary introduced today

- *devarakāḍu* (IAST: devarakāḍu; lit. “sacred grove (Karnataka, S. India)”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today's slice: focus on the part of the activity that reinforces From Ancient to Action.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Day 10 — Project Pitch and Assessment

Module: Indic Ecology — Vasudhaiva Kuṭumbakam · **Band:** Senior · **Time:** 45 min

Learning objective

By the end of this lesson, students will: present project pitches; summative quiz.

Materials

- Whiteboard / chart paper
- Notebook, pen per student
- (See [lesson-plan.md](#) for any day-specific materials)

Vocabulary introduced today

- *sthāvara-jaṅgama* (IAST: *sthāvara-jaṅgama*; lit. “immobile (plants, rocks) and mobile (animals) life”)

Lesson flow

Warm-up (5 min)

- Daily check-in: one-sentence response to “*What’s one thing you remember about yesterday’s lesson?*”
- Hook: introduce a phenomenon or question that motivates today’s concept.

Core (20 min)

1. **State the observable phenomenon first.** Whatever the day’s idea is — a number trick, a celestial pattern, a herb’s identifying feature — start with the thing students can see, hear, or do.
2. **Introduce the Sanskrit term.** Once, in the triple format, then italicised only.
3. **Build the conceptual map.** Connect today’s idea to prior modules (especially Module 2 if applicable) and prior days.
4. **Work an example** with the whole class on the board.
5. **Independent or paired practice** for ~5 minutes.

Activity (15 min)

A scaled version of *Sacred Grove (or Local Ecology) Mapping* (see [activities/](#)). Today’s slice: focus on the part of the activity that reinforces Project Pitch and Assessment.

Wrap-up (5 min)

- One-sentence exit ticket: “*What’s one thing you learned today?*”
- Preview tomorrow.

Homework

(See [homework.md](#) for today’s task.)

Differentiation

- **Extension:** offer one open question or extra problem for fast finishers.
- **Support:** pair students who need scaffolds; offer partial outlines.

Teacher notes

- Citations: at least one quoted verse with citation per module. Day 1 already includes one if applicable; you may add more.
- Connect to prior modules where natural (especially Module 2 *pañcabhūta*).

Writing Prompts

Writing Prompts — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Use these for in-class writing, homework, or assessment. Each prompt is suitable for a 300-word response.

1. In 3–4 sentences, explain Indic Ecology — Vasudhaiva Kuṭumbakam to a friend who hasn’t taken this module.

2. Choose one Sanskrit term from this module. Define it, then describe one observation from your own life that matches.
3. Compare Indic Ecology — Vasudhaiva Kuṭumbakam with a similar concept from another module or from your science textbook. What's similar? Different?
4. Identify ONE claim in this module that you'd want to test. How would you test it?
5. What was the most surprising thing in this module? Why did it surprise you?
6. Pick one source cited in this module. Write 2 sentences explaining why it matters.
7. What is one common misconception about Indic Ecology — Vasudhaiva Kuṭumbakam? Rebut it in 2–3 sentences.
8. Imagine you are teaching this module to a younger student. What would you say first?
9. Read one primary-source passage cited in this module. Translate or paraphrase it; comment on what it adds to your understanding.
10. Identify a contemporary debate about the IKS framework presented here (e.g., is Tirthaji's mathematics genuinely Vedic? are doṣas a valid framework today?). Take a position with two pieces of evidence.

How to use

- Pick **one** prompt per day for the writing-prompts homework.
- Or assign all of them across the module, alternating between in-class and home.
- Senior students may treat any prompt as a 500-word essay topic.

Homework

Homework — Indic Ecology — Vasudhaiva Kuṭumbakam (Senior)

Light daily tasks. Each takes 10–15 minutes.

1. **Day 1 — Vasudhaiva Kuṭumbakam** · Write 2 sentences about today's concept. (~10 min)
2. **Day 2 — Five Elements as Ecological Frame** · Write 2 sentences about today's concept. (~10 min)
3. **Day 3 — Rivers as Sacred** · Write 2 sentences about today's concept. (~10 min)
4. **Day 4 — Sacred Groves** · Write 2 sentences about today's concept. (~10 min)
5. **Day 5 — Yajña — The Ecological Cycle** · Write 2 sentences about today's concept. (~10 min)
6. **Day 6 — Pṛthivī Sūkta (AV 12.1)** · Write 2 sentences about today's concept. (~10 min)
7. **Day 7 — Iṣopaniṣad — Non-Greed Ethic** · Write 2 sentences about today's concept. (~10 min)
8. **Day 8 — Climate Change — The Crisis** · Write 2 sentences about today's concept. (~10 min)
9. **Day 9 — From Ancient to Action** · Write 2 sentences about today's concept. (~10 min)

10. Day 10 — Project Pitch and Assessment · Write 2 sentences about today's concept.
(~10 min)

Notes

- Homework is intended to consolidate what was learned in class, not introduce new content.
- If a child struggles, the parent should consult the teacher rather than push.
- Capstone project days (Module 15) may have lighter daily homework so students can focus on the project.